VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI - 590018

A Project Report on

Design and Development of Fire-Fighting UAV's for the Surveillance of Urban Environments

Submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Engineering
in
Electronics and Communication Engineering
for the **Academic Year: 2023-24**

Submitted by

Adithya T	**1NT20EC006**
C Deepak	**1NT20EC034**
Sai Kiran V	**1NT20EC125**

Under the Guidance of

Dr. Shashidhara K.S
Associate Professor
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NITTE EDUCATION TRUST
NITTE MEENAKSHI INSTITUTE OF TECHNOLOGY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

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Certificate

Certified that the project work titled "**Design and Development of Fire-Fighting UAV's for the Survelliance of Urban Environments"** is carried out by **Adithya T (1NT20EC006)**, **C Deepak(1NT20EC034)** and **Sai Kiran V(1NT20EC125)** bonafide students of Nitte Meenakshi Institute of Technology in partial fulfilment for the award of Bachelor of Engineering in Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi during the academic year 2023-2024. The project report has been approved as it satisfies the academic requirement in respect of the project work prescribed as per the autonomous scheme of Nitte Meenakshi Institute of Technology for the said degree.

Signature of the Guide	**Signature of the HoD**	**Signature of the Principal**
Dr.Shashidhara K S Associate Professor Dept of ECE, NMIT	**Dr. Parameshachari B D** HoD Dept of ECE, NMIT	**Dr. H C Nagaraj** Principal NMIT

External Viva-Voce

Name of Examiners	**Signature with Date**
1.	
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Adithya T	(1NT20EC006)
C Deepak	(1NT20EC034)
Sai Kiran V	(1NT20EC125)

Place: Bengaluru

Date:

Abstract

The project summarizes some of the attempts made in the previous years for the detection and prevention of fire, which was one of the major reasons for the forest fires, and how these methods that were mentioned were advantageous and they also throw a light on some of their disadvantages and how these short-comings can be handled. The use of sensors and on- board cameras in unmanned aerial vehicles [UAV] has grown significantly in the recent age and day when the technology is at an advanced age for the surveillance and monitoring of the forest and man-made fires and also for their prevention. By using Image classification and image processing techniques like SVM and YOLOv8 which are machine learning models, the data sent to the base-station by the UAV captured using the camera will be examined and further actions can be taken. By using these techniques, it is possible to detect areas with possible threat to the nature and population and can be used to take care of extinguishing the fire before it spreads and gives the fire-fighters a proper idea about the surroundings and the overview of the terrain they are dealing with. This prototype can be used to minimize the environmental impact created by the fire while at the same time reducing the damage done to the fire-fighters and leads to the efficient dealing of the fire thereby minimizing the damage caused by the fire and its smoke to the buildings as well as the population around the fire region.

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Chapter 1 Introduction

Fire Fighting is one of the most important occupations of the present day and age, where people and buildings are more prone to accidental fires, also the wildfires in the forests because of the increasing temperature and climate change. But at the same time, the number of people who have chosen firefighting as their profession is slowly and gradually reducing because of the many dangers that are created due to the toxic environment because of many combustible materials, with this profession having many harmful consequences due to smoke, oxygen deficiency, elevated temperatures, poisonous atmospheres, and violent air flows. According to the survey done by the US wildland fire, an average of 70,000 wildfires annually burns around 7 million acres of land and destroy more than 2600 structures leading to huge collateral damage as well as economic loss not only for the people but also for the government [2],[3],[7],[11]. Fire poses a significant threat to life, property, and the environment, necessitating rapid and accurate fire detection for early warning and disaster prevention [24]. These factors combined with the fact that a fire even if a small one if left unchecked can cause huge amounts of damage to the environment as well as the surrounding population makes the fire-fighting an important and a high-priority task. The way of using Humans to carry the firefighting equipment which are used as a traditional way in fighting a forest fire to minimize and extinguish the fire, which usually results in collateral damage to the surrounding environments as well increases the number of casualties caused by the fire as well as the smoke and the chemicals that come along with it and these fires also do harm to the fire fighters up to a certain extent because of the smoke and hazardous elements present in the smoke and fire usually leading to the fighters retiring early either due to the mental and physical fatigue caused by the harrowing scenes at the fires or due to the irreversible damage done to the lungs of these fighters due to the long term exposure to the chemicals and the hazardous elements present in the fire as well as the smoke.

In this project that we have decided to partake in, we attempt to minimize the harm caused to these fire fighters by reducing the time spent by the fighters in the harsh conditions thereby reducing the damage done to their lungs, by making use of one of the fast-developing fields and some of the advanced technologies known as an UAV (Unmanned Aerial Vehicle) which were previously used only for top-secret military applications but are becoming more widely used in the present day and being used in more applications because of the new and upcoming technologies as well as the support and development being done to make the UAV's more compatible to many applications with the fire-fighting being one of the major application.

An UAV which is more renowned as a drone, is an aircraft that is capable of carrying out aerial operations even some of the most dangerous and risky missions and operations without risking any harm to human life as most of the operations are performed autonomously as they can be piloted remotely which is very far from the dangers of the fires. The UAV has shown great advantages in forest fire applications, when it comes detection of the major fire site [1]. Wildfire fighting is a dangerous and a time-sensitive task where each and every second counts, and going into the fire without any proper information or protection/support leads to gambling with one's life which usually takes a turn for the worse [2]. So reaching a fire place as quickly as possible with proper and reliant information is a must for the fire-fighters in order to prevent huge collateral damage, causalities as well as the health of the fire-fighters and this is where an aircraft has its advantage as it can reach the scene of the fire as quickly as possible, which is also easy to manoeuvre over remote areas where it's difficult for the fire-fighting rig to reach [3]. Due to their ability to operate in remote, dangerous and dull situations, the aerial vehicles are now being used in many of the fire-fighting operations like surveying and mapping, data collection from remote places that are inaccessible, rescue operation as well as providing the necessary equipment to the victims [5],[7],[16]. with the help of the cameras and sensors either on-board or using external equipment. Among the many aerial vehicles available for the use in the domain, the Unmanned Aerial Vehicle which is an advancing technology that is capable of carrying out these flight operations without the help of any pilot to control the motion [6]. It is advisable to choose an UAV which is minimalistic in design and as light weight as possible, which is better to have long range of movement, long endurance with flight autonomy, which is capable of communication with the base station either with text, image, video or email when a router is connected to it [4]. Therefore, monitoring multiple areas which are more prone to fire, with the help of UAV's where it is difficult for the fighters to enter will make the situation easier to control when a fire eventually breaks out and can also prevent the situation from escalating the situation by extinguishing the fire in case it's a small one [12].

By using the UAV for detection of the fire by using various machine-learning and deep- learning techniques and communicating this effectively and without any signal loss or to the base station by using the latest image processing techniques. The unique features of the UAV like 3D mobility, low flight altitude, and fast and easy deployment into the fire-site or any other area, makes it a valuable tool for detection and prevention of fire. During the course of this project, we are attempting to introduce an UAV that is capable of recognizing and detecting fire using machine learning and deep-learning algorithms , and also the ability of the UAV to provide real time images and advice the fire fighters during the wild fires as well as the commercial fires. In addition, we are also trying to provide a stable medium of communication that enables the fire fighters to exchange information with the UAV through different means like email, text messaging and voice mail when all the safe and the reliable methods of communication were destroyed as a result of the fire or by the debris caused by the fire. Along with the existing UAV technology for the purpose of detection and transmission of the data from the UAV to the base-station, it is possible to integrate the UAV with heterogenous hardware components with hierarchical soft-ware structures and using some machine learning or deep learning algorithms to more efficiently use this UAV for faster fire-fighting applications. [7],[9]. The combination of the UAV with an on-board camera platforms to increase its computational capabilities, which allows for many operations as mentioned in [5] , [7] , often with higher clarity and better mapping [10]. In order to detect the fire more efficiently it is possible to make use of external features like the colour combination, colour extraction as well as feature extraction from the images and also focus on the combination of static and dynamic characteristics like texture, orientation etc [8]. After detecting the fire, it is also important to have the transmission of this data as quickly as possible as possible without any loss of information as well as the proper signal without any external disturbances or noise added to it as well [14], [15]. In this project, we try to establish a method through which the harm done to the fire-fighters can be significantly reduced and so that the rescue operations can also be completed more easily without much collateral damage and also the harm to the human life by using the UAV for detection of the fire by using various machine-learning and deep-learning techniques and communicating this effectively and without any signal loss or to the base station by using the latest image processing techniques.

For this application a combined application of UAV and IoT devices is pursed [19],[20] making this project a combination of IoT as well as Machine learning algorithms. This communication of data to the fire-fighters can better equip and prepare the fighters in locating and controlling the hotspots of fire as well as provide a better information about the fire and its surroundings, which also leads to the quicker quenching of fire at the source whilst at the same time minimizing the time needed to neutralize the fire and also minimizing the hazardous effects on the fighters. The main focus of this paper is on the coupled task management of detecting the fire and at the same time making use of sensors and using the camera to send and propagate the data [11].

The main focus of this paper is on the coupled task management of detecting the fire and at the same time making use of sensors to send and propagate the data [11].and is therefore It is advisable to choose an UAV which is minimalistic in design and as light weight as possible, which is better to have long range of movement, long endurance with flight autonomy, which is capable of communication with the base station either with through text, image, video or email when a router is connected to it [4].

The aim of this project is to build a prototype of an UAV that is capable of flying through large distances with irregular terrains [3]. Whilst recording the data in the nearby surroundings of the UAV and then propagating/sending the data to the base station using Communication protocols. Doing all this while keeping/ expending only a small amount of battery/power which leads to effective and efficient utilization of the power which thereby increases the flight time of the UAV, and leads to a better usage of the UAV as well as better propagation and increased flight time for the UAV.

In this project, for the image processing techniques we make use of many/ experiment with the three different algorithms and choose the best one that is best suited to the operation and which has a faster and better computation as well as provides the best results in real-time deployment. These features and qualities must be satisfied in order to choose the better algorithm and then the algorithm must then be integrated with the camera on-board so as to propagate the images/information from the camera to the base-station and at the base-station must be classified and identified. The image processing techniques we experimented are Support Vector Machine [SVM], which is a supervised learning algorithm, YOLOv8 which is one of the new technologies being used in most of the present-day image classification and image recognition projects because of its high accuracy and computational abilities, and the HSV [Hue Saturation Value] algorithm which is an unsupervised learning algorithm.

Some of the features that the UAV used needs to have is the ability to carry additional/extra load, be able to propagate to a large height as well as huge distances and should have the ability and the capability to carry such heavy payload without the payload dragging the UAV down or slow down the UAV and the UAV must also be able to carry many sensors as a small camera powered by a small microcomputer , where the microcomputer must be capable of helping the camera in clicking the photos at the regular intervals, converting the photos and must also be able to transmit the photos to the base-station without adding any interference to the data by adapting and utilizing the communication protocols.

Table I. Acronyms used

Acronym	Full Form
UAV	Unmanned Aerial Vehicle
SIFT	Scale-Invariant Feature Transform
MOPS	Multi-Scale Oriented Patches
DCT	Discrete Cosine Transform
EKF	Extended Kalmann Filter
PID	Proportional Integral Derivative
SF-M	Structure From Motion
YOLO	You Only Look Once
R-CNN	Region based Convolution Neural Network
BLDC	Brushless Direct Current Motor
ESC	Electronic Speed Controller

Table I shows some of the most commonly used acronyms used in this project while also expanding and denoting their full form, where some of the acronyms are used to denote the machine-learning algorithms and the common terminology used in building and the features and the properties of the UAV.

1.1 Motivation

UAVs are one of the most advanced and fast developing fields of technology that can be used for many of the real-time applications, so making use of this advancing UAV technology in the field of firefighting and help in faster quenching of the fires and also reducing the time required to analyse the fire while at the same time minimizing the causalities and the harmful effects of the fire to the victims as well as the fire fighters.

One of the major issues to be discussed is that the many UAV’s that are being operated in the present day mainly focusses on the extinguishing of the fire, which usually relies on carrying a huge ball filled with chemicals that can neutralize the fire at the source when the ball is dropped onto the fire [6] but this can be a difficult task in itself because of the mechanism needed to accomplish this task as well as the weight and the infrastructure of the payload mechanism and at the same time the range of the ball that is to be dropped is also very small thereby rendering the method completely redundant and moreover this limits the speed and operability of the UAV, and so does the use of a water spraying mechanism in which the UAV is carrying a load with less water making it ineffective to be used in case of large forest fires , and the use of the water pipes connected to a water source at the site of the operation as used widely and becoming popular in countries like China, the main problem of reaching the site of fire as quickly as possible and examining the situation still remains, as well as the route taken by the fire-fighting rig to reach the spot of the fire.So, we propose to focus more on the detection part as that way, the UAV equipped with a GPS module which is on surveillance duty can be used to not only find the area/ spot with fire [4] [5]. As well provide the path with the least distance and the path without much traffic so that the fire-fighting rig can reach the site of fire as quickly as it can without any interruptions and delay and also at the site can provide the fire-fighters an update on the situation by showing the visuals either through photos transmitted at regular intervals or with the help of live-streaming can provide/show the fire-fighters a clear idea of the situation at the fire-site and gives them a clear picture of the situation inside the fire to giving the fire-fighters more information as to if there are any victims inside the fire and if they are present, gives a clear description of where the victim is , as well as the all the paths of entry so that the fire-fighters can quickly get into the building and rescue the victims without much delay thereby reducing the harm done to the victims as well as the fire-fighters in that particular situation, and also a quick path of exit so the fighters can spend as less time as possible inside the fire which could harm them.

1.2 Organization of the Report

The first chapter of the report talks about the introduction to the project, whereas chapter 2 talks about the research work carried out authors before us, and put forth their findings, applications and their limitations and the Chapter 3 talks about how these limitations can be improved upon and discusses more about the methodologies used to implement the project whereas Chapter 4 talks more about the implementation done to the model and the physical prototype of the model and the Chapter 5 talks about the results obtained from the following implementation done and the final chapter 6 talks about the conclusion of the project and discusses on how the project can be made better and how it can be improved upon.

Chapter 2 Literature Survey

2.1 Background Work

Many authors had previously worked on this topic and came to their own conclusions which were duly noted while their limitations were also mentioned.

The entire literature survey has further been divided into 3 different sections namely:

- MOVEMENT AND PATH CHOSEN BY UAV
- DETECTION OF FIRE
- FIRE PREVENTION

Further discussion on the 3 mentioned topics are shown below:

A. MOVEMENT AND PATH CHOSEN BY UAV

As described in the above section, firefighting is one of the most dangerous and risk-prone jobs in the world that can be made at least somewhat easier with the help of an UAV. The first main job when operating an UAV is the movement of the UAV as well as the path of movement chosen by the UAV that can make it easier for the UAV to reach the base as quickly as possible in the time of duress as well as providing the fire-fighting rig with the easiest and the most optimal path that can help them to reach the site of the fire as soon as possible thereby preventing the situation from getting any worse and the fire quenched as quickly as possible thereby minimizing the threat to life as well as keeping the collateral damage to the minimum[16]. Huy Xuan Pham et al. [2] had proposed the use of a method that uses the combination of sensors and an in- built camera to detect and record the surroundings and make a check-point based method of movement based on the images captured by virtually marking / pinning the locations in the memory, and during the fire would use the discrete grid-method for navigation and movement by avoiding the hotspots that would either damage any of the sensors on-board the UAV and the UAV itself. But one main disadvantage of this method was that after the UAV transverse through the area, the checkpoints would themselves become redundant as they have no further use, whilst at the same time-consuming extra memory which can be used for many useful process like recording the temperature and whatnot. Further improving upon the checkpoint based movement for the UAV, was then upgraded and as shown in the article by Diego A.Saikin et al. [3] proposed a method in which the UAV travelled through an already predefined path that was programmed and set by the programmers and engineers at the base station. The path was chosen by surveillance done by the base station and then uploading the path into the memory of the UAV with the help of memory storing modules present in the UAV, while choosing the path to different locations and test-driving the UAV to ensure the efficiency of the UAV and changing the path in case the efficiency and effectivity was less in the chosen path. One major disadvantage that was encountered during this process was that since the path was already predefined, the base-station had no control over the propagation of the UAV, which meant that if the UAV malfunctions and goes rogue during the travelling phase, it becomes difficult to search and retrieve the UAV and also becomes difficult to reprogram the UAV mid-journey and also during the fire, since the path is pre-defined there is a high-danger that the UAV may enter into the fire thereby damaging the sensors present and the UAV itself. Moving on, from the check - point based navigation system that had many disadvantages, so Abdulla Al-Kaff et al. [5] proposed the use of a real-time flight control by the pilot, by giving him the visuals on the base station transferred through the camera present on-board the UAV. And the pilot would be able to mainour the UAV based on the input he received from the cameras on-board the UAV. What this meant was that the UAV path was not defined and the path was controlled in real-time, which meant that the UAV could change directions mid- journey based on the circumstances. but using control from the pilot meant that the UAV completely loses its autonomous operation, but also in case of longer distances when the distance between the UAV and the base station is very far, due to the latency produced it becomes difficult to navigate the UAV properly as there will be a delay in communication between base station and the UAV. As the method suggested and used in [5] had many disadvantages, Abd. Hafiz Zakaria et al. [6] proposed the use of one of the most promising and developing technology known as PID technology. This technology is a program that is used for controlling the UAV based on the images captured by the sensors present on-board the UAV and then drives the motors accordingly via the motor controller, so thereby giving the autonomous feature to the drone, and also the path can be tracked and modified by the UAV itself mid-journey in case some obstacle or object comes in the path of propagation of the UAV. Now, moving on one of the other technologies that can be used for UAV propagation other than PID technology is an RTK-based navigation system as proposed by Hailong Qin, et al. [7] who suggested the use of this RTKbased system rather than the PID controller. The RTK-based communication is also known as Real-Time Kinematics, which consists of a GPS receiver and tracker that is used to pin-point the location of the UAV or the fire scene with a positional accuracy of 1cm. It's because of this accuracy parameter that many UAV developers are considering switching to this RTK-based navigation system over all the other systems present in the market. This method too has its disadvantages, but at the same time minimizes or completely Acronym Full Form UAV Unmanned Ariel Vehicle SIFT Scale-Invariant Feature Transform MOPS Multi-scale Oriented-Patches DCT Discrete Cosine Transform EKF Extended Kalman Filter PID Proportional Integral Derivative SF-M Structure From-Motion YOLO You Only Look Once R-CNN Region based-Convolution Neural Network neutralizes some of the major issues that were presented when using the other methods.

B. DETECTION OF FIRE

The first part of section II talked about the movement and the path taken by the UAV for propagation. Now let us talk the second important operation which is detection of the fire by the UAV when it reaches the fire site. Now there are many methods proposed which can do this operation, but one of the most commonly and widely approaches for this operation make use of machine- learning and deep-learning process, here are some of the researchers that have used this method for detection of fire. Some of the most common types of features used in machine learning approaches are as shown below:

• Smoke

• Temperature of the surroundings

• Intensity of the fire

Starting off with Abdulla Al-Kall et al. [5] suggested the use of an on-board monocular camera for the image capturing that can then be sent to the base station where these images can be extracted and classified based on the features present in the image and also proposed the use of an on-board camera which meant that the UAV would be very light thereby enabling it to flying to a longer distance without using too much of the battery. This paper proposed three methods that can be used for the analysis of the image they are:

(1) Combination of SIFT and MOPS with corners enhancement where the features are extracted using MOPS.

(2) Using Hough transform for indoor environments but the limitation was that it was only applicable for the indoor corridors and stair areas

(3) Using DCT method to the input images which removes the background and then using IDCT to obtain the main centre part of the image where the fire is raging. Also, the relative distance for transmission of data is measured using EKF.

The algorithm used here is SIFT which mimics the human behaviour, and after the detection of the fire, brute force algorithm is used for further dissecting the image for more visual clarification. Coming to the next method proposed by Hai long Qin et al. [9] had suggested in one of his works that the use of a combination of sensors and thermal imaging would not only make the detection process much easier but also at the same time keep it more efficient , as the sensors would detect and send the images to the base station, and during the night time , when the sensors had little to no effect because of the darkness, the thermal imaging would become active and transmit the images for surveillance, thereby making sure that he UAV can operate both in day and night time without encountering much of the problems and at the same time transmitting quality and efficient data to the base station that can be decoded and classified easily. Chen Zhang et al. [11] proposed an approach where the task planning approach was used and an action-based approach was considered. The images were captured with the help of an on-board camera and then sent into the base station where the features of this image were extracted and then classified based on Dublin’s curve. This method also made use of the re-enforcement method of learning where the system gets rewarded every time it makes a proper and correct decision. this made sure that the machine learning model never repeated the same mistake as it learnt and adapted from its mistake. Therefore, the errors and the wrong information encountered in this method were minimum to none. Liyana Adilla Binti Burhanuddin et al. [12] suggested the use of multiple UAV’s for communication and transmission of data, where they talked about using uplink between two UAV’s to propagate to large and longer distances and used most the modern communication techniques for this purpose, Game theory techniques were preferred and the DRL transmission technology was used. As Xiwen chen, et al. [15] has mentioned:

1. To use the deep learning-based method, a series of Convoluted Neural Networks which are used for various tasks like fire detection, object detection [5]. where the classification is performed by extracting the high-level features and comparing their properties

2. In this method, the data set given to the database of the ground station where the fire is controlled and minimized by using the data (ex: velocity of the wind etc..). The UAV will classify and alert the ground station based on this data. A. YOLO Algorithm:

Guangyuan Zhao at el. [21] has proposed the usage of ghost model algorithm that reduces the dimensions of the target object and reduces the no. of parameters and convolution computation that will improve and make faster when compared to the traditional fire-based detection system which has high latency. They also proposed the use of a deep learning algorithm in real-time which increases the accuracy when compared to the old traditional classification mechanism. They concluded their paper by suggesting the use of the YOLOv5 Algorithm which was faster as well as more accurate. One major disadvantage of using this method was that it becomes difficult to implement it on the embedded devices because of their limited memory and computation power. Wanbo luo [22] has suggested the use of supervised learning algorithm that produces a lot of labels on the provided data-set that is used to train the model by identify fire as well as improve accuracy. He further delved deeper into the process as he goes onto explain the two different stages of the YOLOv5 algorithm. First stage being the training of the model from the pre-defined data-set and define the machine-learning algorithm The second stage is the deployment of the training model to test the accuracy and prediction of the model. He also summarized by saying that the YOLOv5 learning algorithm when compared to the traditional detection is faster to the user and at the same time has less latency because of the model already being pretrained, whilst giving better accuracy and prediction for a faster detection in the real-life application Marwa Jmal at el. [23] has proposed the usage of a hybrid model that combines the YOLOv5 and U-net architecture for the detection and segmentation of the fire. He has also implied that this helps in the early detection of the fire and prevents the fire from creating any additional damages. The YOLOv5 Algorithm is a single-stage object detection algorithm that has 3 important parts: Model Backbone: This part is used to extract the important features form the data collected using the camera and can be used as a backbone to extract the informative features from the input fire image. Model Neck: In this step, the object scattering and object scaling happens when the same object has different scales and sizes. Model Head: This is the final detection part that YOLO employs to create anchor boxes around the features that have been extracted and have been important and develops the final output for the input image namely, boundary boxes, with the confidence score in the prediction with a class score. Liancheng Su et al. [24] suggested the use of a YOLOv5 algorithm that focuses on enhancing real-time flame and smoke detection using the YOLOv5s network. It addresses the challenges of slow calculation speed, low accuracy, and high false detection rates in flame recognition algorithms based on YOLOv5s. The proposed method, FC-YOLO, integrates FLAnet and CoordConv to improve detection efficiency and accuracy. By leveraging data set preprocessing techniques and enhancements, the improved model achieves a higher mean Average Precision (mAP) value of 73.6%, indicating a 3.5% increase in accuracy compared to the original model. The enhanced detection algorithm aims to boost fire monitoring efficiency, reduce fire-related losses, and hold practical significance in enhancing fire safety measures. Since the

YOLOv5 was a bit out-dated when it comes to detection there is a more updated and a better version as mentioned in his paper by Xinchen Zou [25] who introduces an enhanced flame detection algorithm based on YOLOv7 to address challenges like complex flame features and susceptibility to environmental interference. By leveraging the YOLOv7 target detection algorithm, the study aims to enhance flame detection accuracy in real-time scenarios. And mentioned that the use of techniques such as K-Means clustering for adjusting priori frame dimensions, loss function of MPDIoU to reduce unnecessary feature learning, ECA attention mechanism for spatial perception enhancement, and DSConv for speed and memory optimization are incorporated to improve flame smoke detection accuracy. Ning Jiang et al.[26] had also proposed the use of a lightweight and reliable flame detection algorithm based on the YOLOv3 algorithm, enhancing its real-time performance and credibility where he also improved the backbone of the YOLOv3 algorithm by using depth wise separable convolution to streamline the model and boost real-time performance and also added parameters like an attention mechanism and fusion features detection head to enhance the expressiveness of features and improve target prediction efficiency. He also trained 3 different variations of the same model The YOLOv3 model, YOLOv3-tiny model, and an improved YOLOv3 flame detection model using the dataset and then compared them with each other to find out which algorithm gives him the best efficiency.

C. FIRE PREVENTION

The first part of section II talked about the movement and the path taken by the UAV for propagation and the second part talked about the fire detection. Now let us discuss about the measures that are taken in order to prevent the fire from escalating the fire, Diego A.Saikin et al. [3] had suggested using water reagent to quench the fire which almost negate all the disadvantages of the previous method since water is less heavy in the contrast, the water can be releases at any angle possible without fearing any damage to the UAV. But one major disadvantage of this method was that since the reagent used to quench the fire was water, it had to be done to the ground zero of the fire, as because the re- agent used is water it gets evaporated when released from a height or may start to drift because of the wind. Zhan-Ke Li et al. [13] talked about the idea of having 4 sand bombs instead of 2 as previously mentioned in [6] and also about upgrading the hexacopter to a eight motored UAV that worked on LI-ion battery thereby making it eco-friendly and at the same increases the re- usability of the battery as the LI-ion battery can be charged. The primary findings of the research done on the man-made as well as the forest fires as well as the related solutions are shown in the Table II.

Table II. Main findings regarding research conducted

References	Findings and Observations
Abdulla Al-Kall et al. [5]	1) Object detection was made possible by machine learning techniques. 2) Accurate prediction using Sift technique. 3) Detection using brute force.
Luis F.Romero et al. [17]	1) Suggests the use of check-points for UAV to provide very efficient communication. 2) Quick and early response from the base-stations is possible using In-built camera in the UAV. 3) Optimal Flight Paths are calculated using Centroidal Voronoi Tessellation method.
Huy Xuan Pham et al. [2]	1) Uses a combination of camera and sensor. 2) Uses check-point based method of movement. 3) Voronoi Partition is considered.
Hailong Qin et al. [9]	1) RTK-navigation method is used. 2) Fire detection using sensors and thermal imaging.
Abd. Hafiz Zakaria et al. [6]	1) Using PID technology for transmission. 2) Using sandbags for effective quenching of fire. 3) Hexa-copter is used in place of the quadcopter.
E.A Yfanis et al. [4]	1) Uses LIDAR for propagation and recognition. 2) Solar powered UAV is used which is eco-friendly. 3) CNN algorithm is used for efficient pattern recognition.
Samuel Oswald et al. [10]	1) Uses SF-M technique for propagation. 2) MAPEO structure is used. 3) Data processing is done using AWS. 4) R-CNN is used for data classification.
Hanh Dang-Ngoc et al.[8]	1) Colour-based detection methods are used. 2) Pixels directly extracted and used for classification. 3) Optical flow algorithm used to examine and extract the features.
Jeyavel J et al. [18]	1) Linear Quadratic Estimator (LQE) and EXF are used. 2) Lyapunov's stabilization process is used.

2.2 Open Issues and Challenges

From all the researches that were conducted, the one major challenges that most of the researchers met with was the implementation of the ML algorithm that could easily detect and designate the images are fire, but some of the modern day researchers presented the idea of use of a new and improved algorithm known as the YOLOv8 algorithm and as Yike Wang et al.[27] has suggested that the a forest fire detection strategy that involves building a system with pre-processed and pre-trained datasets, receiving images from UAVs, and using algorithms to calculate the fire occurrence rate, aiding in speeding up rescue operations. He also conducts an in-depth analysis of YOLO series technology in image recognition, specifically comparing YOLOv8, YOLOv7, and YOLOv5 models based on the accuracy of fire detection and model training speed. YOLOv8 is found to have the best accuracy and a good balance between accuracy and computational speed, making it the chosen algorithm for the detection model. He also proposed that a forest fire detection system needs to have and must involve UAVs locating fires, capturing scene imagery, transmitting data wirelessly to a processing platform, and utilizing a pre-processed dataset with an appropriate algorithm model to estimate fire incidence confidence and relay outputs for prompt rescue efforts. He also talks about the experimental results and comparisons, focusing on the performance metrics such as Mean Average Precision (mAP) across different YOLO models and epochs, showcasing the effectiveness of the YOLOv8n model in achieving high mAP values. similarly, Xin Geng et al. [28] he introduces the YOLO-SF algorithm, which combines instance segmentation technology with the YOLOv7-Tiny object detection algorithm to enhance accuracy and detection capabilities much the detection and segmentation much easier. He also proposed the use of a fire segmentation dataset (FSD) which was created by gathering images containing fire and non-fire elements which were used for training the model. The segmentation detection head of YOLOR is utilized to improve model accuracy and detail expression. He measured the algorithm's performance by evaluating metrics such as Precision, Recall, mAP, FPS, and many such Parameters for both Box and Mask outputs. whereas Wentao Xiong [29] describes a method for obtaining similarity features between adjacent frames of binary images to identify suspicious flame areas, marking pixels with a median value of 1 to define the flame area where he made use of a pattern recognition for real-time fire identification and monitoring and used a weighted average method for Gray processing, averaging the R, G, and B components with different weights to extract edges, encode them based on shape and curvature, and derive feature quantities for fire judgment which included the extraction of features/data from the images which were important and then discarding the remaining information. Zongxin Wang et al. [30] proposes a method that combines a classification model and an object model for forest fire recognition, leading to the development of a fire warning system where he trained and tested on the YOLO algorithm and then a decision tree method was employed for joint decision-making based on classification and detection results. He also built a forest fire early warning system with separate front-end and back-end components is developed. Unlike some of the earlier researchers, he made use of Various real-world forest fire data sets, including images of flames, smoke, and forestry scenes. Mily Lal et al.[31] introduces a novel fire detection system based on Convolutional Neural Network (CNN) technology, showcasing remarkable accuracy and efficiency in detecting flames of various scales, including small-scale flames, in diverse scenarios which were not possible using the older ML algorithms where she Integrated colour and stir features of bank pixels enhances the system's capacity to identify fires, significantly improving fire safety measures and enabling prompt alerts during fire accidents. Xianghong Xue et al. [32] proposes a UAV-based forest fire detection method utilizing dual-light vision where visible light images in YCrCb colour space and infrared (IR) grayscale images are used to detect flames. Visible light images in HSV colour space are utilized to detect smoke, enabling potential early fire detection. The method involves aligning visible and infrared images to obtain mask images of the detection objects, visualizing the results. He also combined Colour features and infrared information to design flame and smoke threshold segmentation rules for fire detection tasks. M Suguna et al. [33] focuses more on early thermal forest fire detection using UAVs and saliency maps. Special maps called the Saliency maps, are generated using the BAS-Net architecture, which includes a predict-refine architecture and a hybrid loss for accurate image segmentation. All these papers could be used as a reference for many of the open challenges that were and are being encountered in this project. But one way this project could be improved is that the use of a hybrid ML algorithm that can train itself on the go but taking images straight from the environment and train itself on the go using a ML method known as the Continuous Learning Algorithm can be deployed and the integration of this algorithm with the already existing algorithms like the YOLOv8 or many more could be used in order to get faster computation results and better accuracy and efficiency as well as the use of small fire-extinguisher on the underneath of the UAV could be used in order to provide the UAV a better progress and can improve the fidelity and the movement of the UAV as well as provide the camera present on the UAV with a better visual and more clarity on the surroundings which has fire to it making the images more clear which leads to faster computation and better results as well as provide the fighter at the base station a better understanding of the situation.

2.3 Problem Definition

To design and develop an Unmanned Aerial Vehicle which can be used to explore fire prone area which is dangerous for humans to enter and analyse the situation. Due to the maneuvering capability of a UAV, it is more efficient to send these UAV to fire prone areas which can perform better analysis than a well-trained fire-fighter who has to risk his life for this purpose.

The implementation of these Fire-fighting UAV has been vastly observed in Forest Fires but very less research and implementation are done in urban environments due to the terrain complexity in these urban areas. Hence, in this project we will explore various method to detect the fire, send the images of site of fire source and help fire fighters to plan accordingly with the information received from the UAV.

2.4 Objectives

- This project aims to design a machine learning algorithm to detect fire from the image captured by the on-board camera present in the UAV using Raspberry Pi. That is to create a sophisticated machine learning algorithm capable of identifying instances of fire within images captured by an on-board camera mounted on a UAV (Unmanned Aerial Vehicle). Leveraging the computational capabilities of the Raspberry Pi micro-controller, the algorithm will analyse visual data in real-time to accurately detect the presence of fires. This algorithm will employ advanced image processing techniques, possibly including convolutional neural networks (CNNs) or other deep learning architectures, to efficiently and reliably identify fire patterns and distinguish them from background elements.
- This project also involves the design and construction of a specialized UAV equipped with firefighting capabilities. The UAV will be engineered to carry a payload of up to 1 kg, which may include fire suppressants, sensors, or other firefighting equipment. The Raspberry Pi micro-controller will serve as the central processing unit for the UAV, managing flight controls, data processing, and communication with ground stations or other networked devices. Additionally, the on-board camera will provide crucial visual information for both navigation and fire detection purposes. Through careful integration of hardware components and software systems, this UAV will be capable of autonomously detecting and responding to fire incidents, potentially mitigating risks and aiding in firefighting efforts across various environments.

2.5 Scope of the Work

One key aspect of the envisioned enhancement is the implementation of autonomy, enabling the UAV to operate independently in dynamically changing environments. Through advanced algorithms and decision-making processes, the UAV will not only detect fires but also analyze the situation in real-time to determine the optimal course of action. This includes assessing the severity of the fire, identifying its source, and formulating an effective strategy for extinguishment.

Moreover, the integration of transceivers and communication systems will enable seamless interaction between the UAV and external stakeholders, such as emergency response teams and urban infrastructure networks. This two-way communication facilitates the exchange of critical information, allowing for coordinated firefighting efforts and improved situational awareness.

Furthermore, the UAV's capabilities can be expanded to include proactive measures aimed at preventing the spread of fires. By deploying preemptive extinguishment techniques at the source during the initial stages of a fire outbreak, the UAV can help contain the incident before it escalates into a larger conflagration. This proactive approach not only minimizes property damage and loss but also enhances overall safety for residents and responders alike.

Chapter 3 Design Approach and Methodology

3.1 Image Processing

3.1.1 Models used along with their description

In this project, as mentioned in Chapter 1 we have used/experimented with multiple Algorithms in order to find the model that works good with the real-time data, produces accurate results with the least computation time.

The algorithms used for this purpose are:

- CNN Algorithm
- SVM Algorithm
- YOLOv8 Algorithm
- HSV Algorithm.

Some of the basic features and the set-up of these algorithms are as shown below:

3.1.1.1 CNN Algorithm

The steps involved in training the model/algorithm and the different layers used are as shown below:

- **Data Preparation:** The code assumes that the dataset is organized into training and testing folders (forest_fire/Training and Validation and forest_fire/Testing), each containing subfolders for fire and non-fire images. The split folders package is used to split the data into training, validation, and testing sets. Image-Data-Generator is utilized for data augmentation and preprocessing.

- **Model Architecture:** The CNN model consists of several Conv2D layers with 'relu' activation followed by MaxPooling2D layers for down sampling. It ends with Dense layers for classification, using 'sigmoid' activation for binary classification (fire or no fire).
- **Training:** The model is compiled with the Adam optimizer and binary cross-entropy loss function. The fit() function trains the model using the training dataset for a specified number of epochs while validating on the test dataset.

- **Prediction and Evaluation:** After training, the model is used to predict classes for the test dataset. Predictions are rounded to either 0 (indicating fire) or 1 (indicating no fire). The code plots the training and validation loss as well as accuracy across epochs using Matplotlib.
- **Image Prediction Function:** The predictImage() function is defined to predict the class (fire or no fire) of a given image using the trained model. It loads the image, preprocesses it, predicts the class using the model, and displays the image with a label indicating the predicted class.

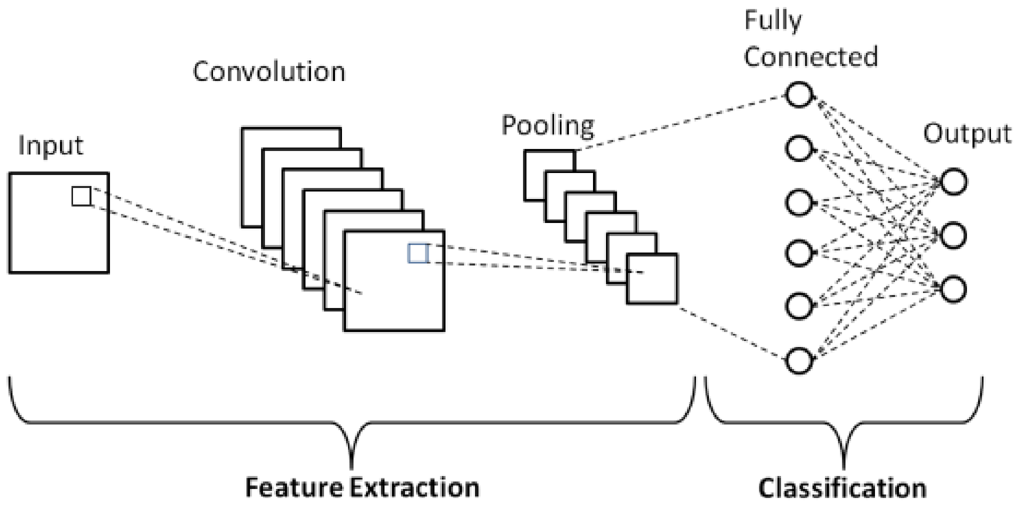

Fig 3.1: Schematic Diagram of a basic CNN Network

The basic schematic diagram of a basic CNN network is as shown Figure 3.1 where after the input is fed into the model it has 2 layers namely convolution and polling layer, where the convolution layer is used to extract the important features from the main photo and discard the remaining useless or redundant information thereby reducing the size of the image [31] while the pooling layer is used reduce the dimensions of the features so that the model training can happen more easily and at a pace that is quicker than usual.

But the one main disadvantage of using this layer is that the model requires a large amount of dataset to train accurately which usually takes a lot of time to train and thereby deployment of the model as well will be delayed because of this and the algorithm takes a lot of time in order to learn the patterns of the images effectively which thereby means that the accuracy of the model at the starting stages is going to be low which may result in false results being displayed by the model, so the next model to be examined was the SVM [Support Vector Machine] algorithm.

3.1.1.2 Support Vector Machine [SVM] Algorithm

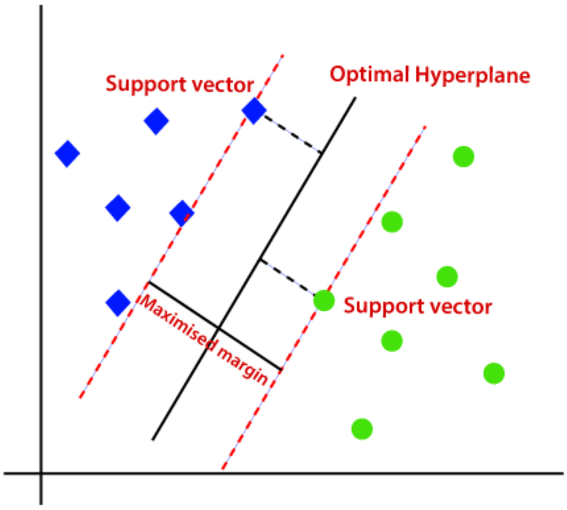

Fig 3.2: SVM Algorithm

Early and accurate fire detection is crucial for minimizing damage and ensuring safety. Machine learning algorithms offer promising solutions for automated fire detection in various applications. This report explores the use of Support Vector Machines (SVMs) for fire image classification. SVMs are supervised learning models adept at classification tasks. They work by finding a hyperplane in a high-dimensional feature space that best separates the data points belonging to different classes. In fire detection, the classes would be fire and non-fire. Figure 3.2 shows the SVM Hyperplane Visualization (Imagine a 2D plane here separating fire and non-fire data points). The features of SVM Algorithm is given below:

- **Data Points:** Represented by circles and squares, these depict data points from the training set. Circles could represent features extracted from fire images, while squares represent non-fire features.
- **Hyperplane:** The solid line represents the decision boundary learned by the SVM. It maximizes the margin between the two classes.
- **Support Vectors:** Data points closest to the hyperplane on either side are called support vectors (shown as larger circles and squares). These points are crucial for defining the hyperplane.

Fire detection with SVMs is a multi-step process that leverages the algorithm's strength in learning complex patterns from image data. The first crucial step involves gathering a comprehensive dataset of images encompassing a wide range of fire scenarios (small flames, large infernos), diverse backgrounds (forests, buildings), and varying lighting conditions (daytime, nighttime). The more diverse the data, the better the SVM can generalize and handle real-world variations. From the images, relevant features that distinguish fire from non-fire are extracted in a process called feature engineering. Common features include color (algorithms identify the presence of red, orange, and yellow hues characteristic of flames), intensity (bright pixels are targeted as potential indicators of fire), and texture (flickering patterns are analyzed to differentiate fire's dynamic nature from static objects). Depending on the application, additional features like temperature data from sensors or smoke detection algorithms can be incorporated for a more comprehensive approach. The collected images are then labeled as fire or non-fire. This labeled data is fed into the SVM, which essentially learns to draw a hyperplane in a high-dimensional feature space. This hyperplane acts as a decision boundary, effectively separating the fire and non-fire data points based on the extracted features. The SVM specifically focuses on identifying the data points closest to the hyperplane on either side (called support vectors) as these are most crucial for defining the optimal separation boundary. Once trained, the SVM is ready to classify new images. When a new image is presented, the SVM analyzes the extracted features and compares them to the learned hyperplane. Based on this comparison, the SVM classifies the new image as fire or non-fire.

3.1.1.3 YOLO Algorithm

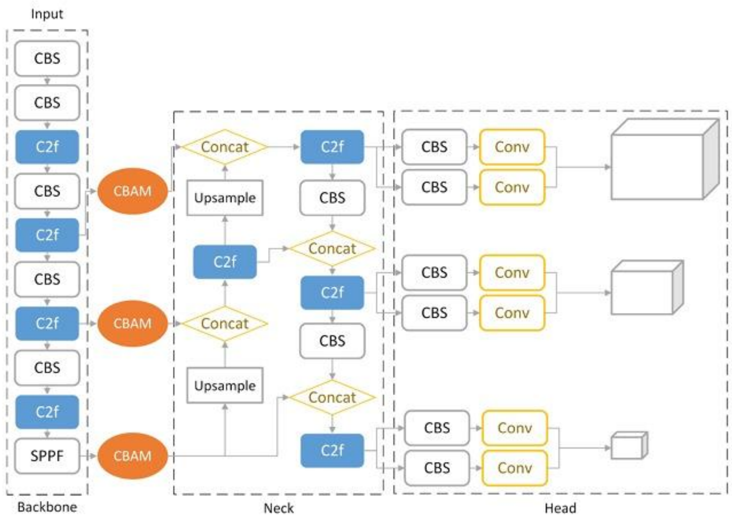

Fig 3.3: YOLOV8 with CBAM Module

The Figure 3.3 refers to the structure of the YOLOv8-CBAM model, which is a combination of the YOLOv8 algorithm and the Convolutional Block Attention Module

(CBAM). YOLOv8: You Only Look Once version 8, a real-time object detection algorithm that divides the image into a grid and predicts bounding boxes and class probabilities for each grid cell. CBAM: Convolutional Block Attention Module, a module that enhances feature extraction capabilities by focusing on important regions of an image. The YOLOv8-CBAM structure combines the efficiency of YOLOv8 for object recognition with the feature enhancement capabilities of CBAM, resulting in improved accuracy for fire and smoke detection tasks. The importance of this algorithm is given below:

- Fire and smoke pose significant threats due to their rapid spread and destructive nature, especially in areas like residential zones, airports, and forests, necessitating swift control and detection.
- Traditional sensor-based fire detection methods are limited in larger settings, leading to delays in extinguishing fires, prompting the need for vision-based detection methods.
- Vision-based fire detection offers advantages like rapid response, extensive coverage, and environmental robustness, making it increasingly popular.
- Research on fire and smoke detection has shifted towards video image detection algorithms, with deep learning-based approaches showing superior performance over traditional methods.
- The YOLOv8-CBAM algorithm, integrating the Convolutional Block Attention Module (CBAM) with YOLOv8, enhances smoke and fire detection accuracy by 2.3%, surpassing other state-of-the-art methods.

3.1.1.4 HSV Algorithm

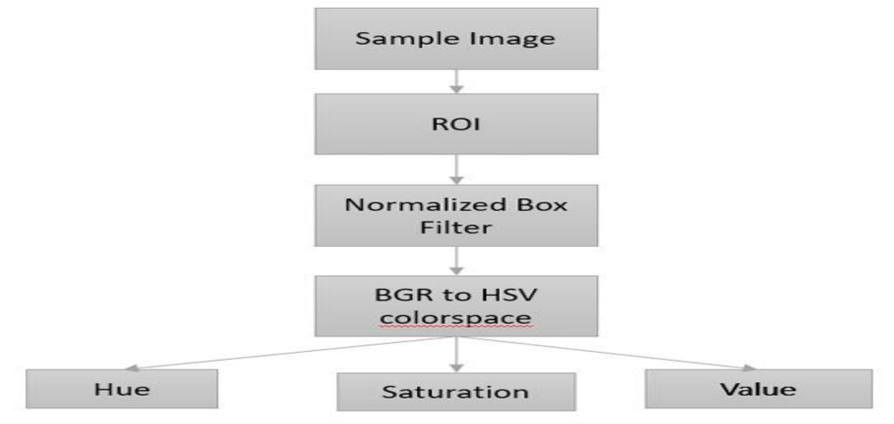

Fig 3.4: HSV Algorithm

The Figure 3.4 refers to an algorithm used for image processing utilizing OpenCV to calculate the HSV colour space coordinates value. It shows an image processed using OpenCV to calculate HSV colour space coordinates in a fire image sample. The components used are OpenCV which is a library programming feature for Computer Vision used for extracting information from images in real-time. HSV colour space is a colour model that represents colours in terms of hue, saturation, and value. This algorithm is crucial for calculating the HSV colour space coordinates value, which is essential for colorimetry analysis in measuring colour concentration in water. By utilizing OpenCV for image processing, researchers can accurately determine the colour properties of water samples, aiding in water quality assessment and pollutant concentration measurement. The implementation of this algorithm is given below:

- HSV colour space coordinates are essential for measuring the sample concentration accurately, as they provide a standardized way to represent colours in an image.
- The OpenCV algorithm used in the image processing converts the image into a digital format, allowing for precise analysis of colour.
- By following step-by-step instructions in the OpenCV code, the computer can calculate the HSV colour space coordinates efficiently, enabling accurate measurement of fire source in the image.
- Overall, Figure 3.5 demonstrates the practical application of OpenCV in processing images for fire detection.

3.2 Datasets used

In this research, a new dataset called FIRE Dataset which was created by carefully selecting and pre-processing the images present in the dataset[26] and these images having fire and no fire were taken from a publicly accessible dataset repository called Kaggle, in which the dataset being used was created and contributed by the user Ahmed Gamaleldin which originally containing 755 outdoor fire images with some of them containing heavy smoke so that the model can train more accurately, the dataset also contained 244 non fire images which were also used to train the model with the dataset mainly comprising of fires in the forest area and in areas where there was abundant greenery. The image showing the dataset used is shown below in Figure 3.5.

The algorithm used this dataset in which the dataset was split into 3 sections for training, testing and validating the model. spilt was done in such a way that 75% of the data was used for training the model and the remaining 12.5% was equally divided as the testing and the validation data with more information about the dataset is given in the below Table III.

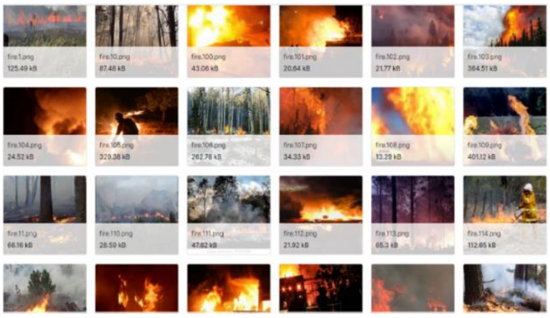

Fig 3.5.Dataset used for training the model

TABLE III. Sample Composition of the Fire Dataset

Sample Category	Number of training sets	Number of validation sets	Number of test cases	Total
Fire	565	92	98	755
No Fire	183	30	31	244

3.3 Workflow of the UAV

3.3.1 Overflow of the Working of UAV

Flow chart 1:Overall working of the UAV and working of the different stages and also the data produced in those stages:

The Figure 3.6 shows the glimpse and the overall working stage of the UAV in all its different stages and the different stages and the operations that take place in those stages.

- The first stage is the basic and the most important stage when it comes to operating the UAV as this involves the selection of the pilot who's skilled enough in order to manoeuvre the UAV over all the irregular terrains and also defining the important hotspots/zones which are prone to fire and setup regular surveillance over those areas and then setting up the output resolution for the transfer of images.
- The second stage talks about the FLIGHT process that involves flying the UAV whilst measuring the surrounding parameters such as temperature and heat by using the sensors on-board the UAV as well as the images using the camera present on-board, at the end of this stage the raw data from the

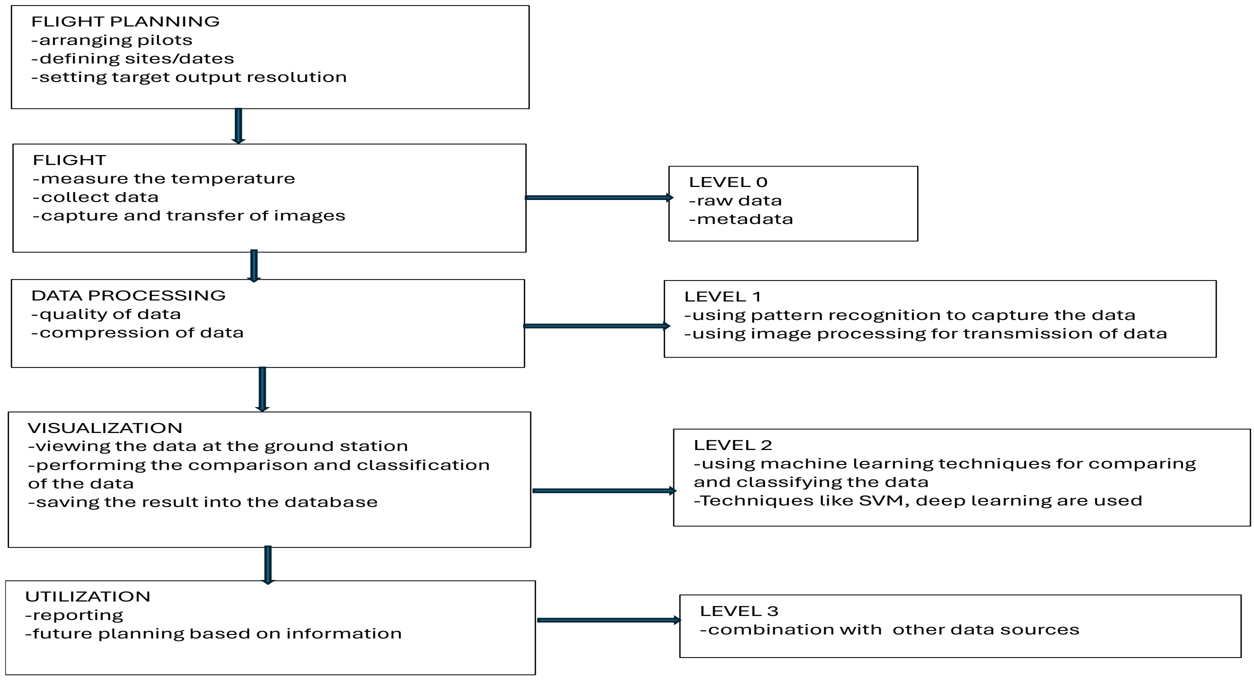

Fig 3.6 overview of the working of the UAV

measurements is extracted and the meta-data is prepared in order to be transmitted to the base-station.

- The next stage following the Flight an the data collection comes the data processing stage in which the data pre-processing takes place and the quality od the data is improved when it comes to the images that are being captured by the UAV and at the same time as the data is being improved, it is also being compressed in order to be able to transmit it more freely to the base-station without much interference. The internal processes taking place in this stage include pattern recognition based on the situation and then the transmission of the data being done using the communication protocols.

- The fourth stage is the heart of the operation process where the data that is being transmitted to the base station is processed using certain machine learning algorithms for the classification and the processing of data after which the data will then be saved into the database which will then be used for classification purposes.
- The fifth and the final stage talks about the results of the analysed reports and the measurements that are to be taken into account while considering the data that is obtained after the image processing takes place and the further action that needs to be taken by the officer based on the data obtained.

3.3.2 Transmission of data from UAV to base station

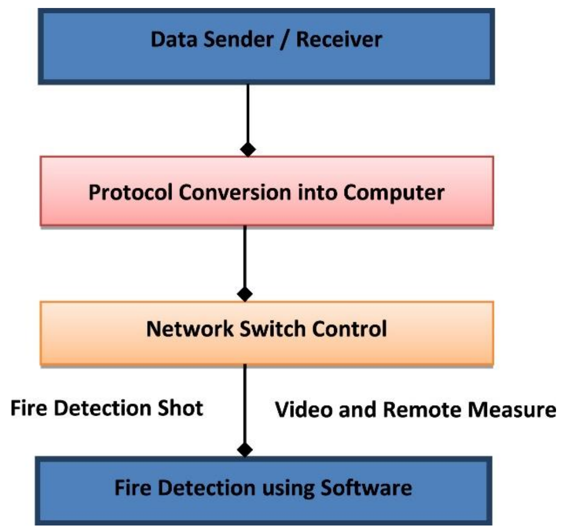

Fig 3.7 transmission of data from drone to base station

As mentioned in the above figure, we need a proper communication link between the drone and the base station when all the communication channels breakdown, the flow chart shown in figure 3.7 gives a brief description as to how it would happen.

Beginning from the sensor present on the drone will capture the information from the area to be inspected and then coverts and encrypts the data for safe and accurate communication of data which is then received by the base station where it the original image will be received and analyzed.

3.3.3 Representation of Data Transmission

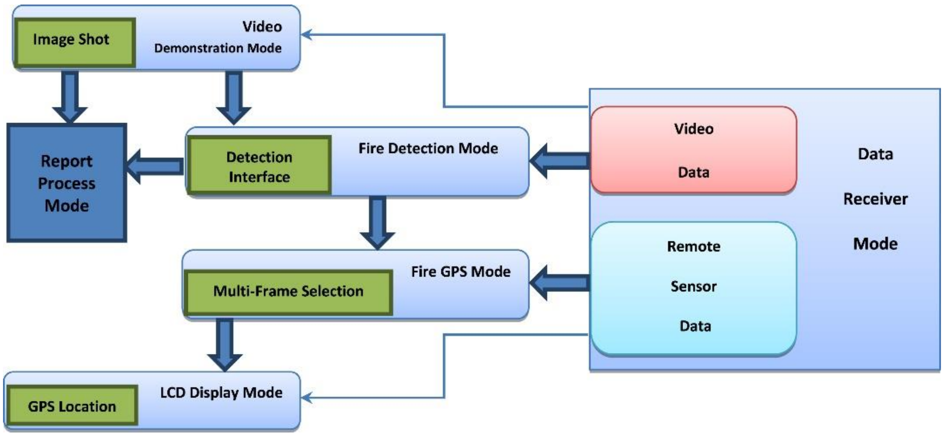

Fig 3.8 in-depth representation of data transmission

The above Figure 3.8 shows the in-depth and the complete process that takes place from when the drone detects and clicks the image to the data transmission till the time when it reaches the base station with all the required specifications.

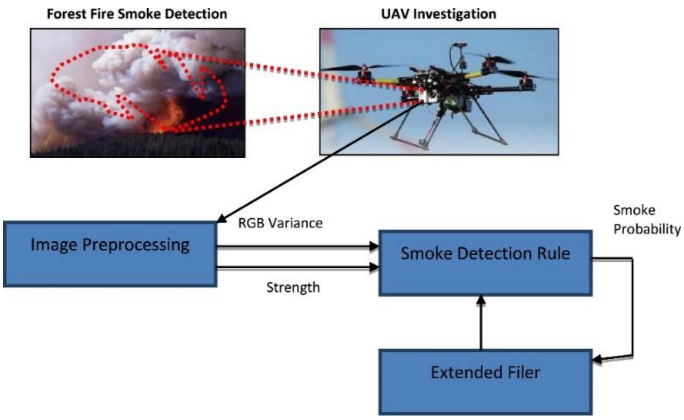

Fig 3.9 basic working of drone for transmission of data

The above Figure 3.9 shows the operation of the UAV when the transmission of data actually takes place starting from when the image gets taken and encapsulated to the image pre-processing stage when the smoke-detection rule is applied to check the intensity of the fire and how hazardous the situation is and then uses the images present in the database to do this operation, and then uses the EKF to further extract the data and then transmits the samples back to the base station for further analysis.

Chapter 4 Implementation Details

The implementation of the entire project has been divided into 2 brief sections namely the hardware implementation and the software implementation which is explained briefly below:

4.1 Hardware Set-up

The hardware set-up can be further split up into the implementation of image processing and image classification as well as the set-up for the UAV.

4.1.1 Hardware set-up for image processing

For the image processing and image classification part we have decided to use the raspberry pi micro-computer unit along with a pi camera module because the raspberry pi having wi-fi and Bluetooth capabilities is a big advantage that can be capitalized upon for an even accurate and better data transmission when preferred to the other micro-controller modules like Arduino Uno. And the basic set-up required for the image capturing and image processing is as shown below.

- **Acquire Raspberry Pi and Camera Module:** Obtain a Raspberry Pi board (Raspberry Pi B Model 4) and a compatible camera module (e.g., Raspberry Pi Camera Module V2). Ensure that the camera module is connected to the Raspberry Pi correctly following the manufacturer's instructions.
- **Connect the Raspberry Pi:** Connect the necessary peripherals to the Raspberry Pi (keyboard, mouse, monitor) and power it using a reliable power source (micro-USB power supply).
- **Check Raspberry Pi Configuration:** Boot up the Raspberry Pi and ensure that the camera interface is enabled. This can be done through the Raspberry Pi Configuration interface or by running **sudo raspi-config** in the terminal and enabling the camera interface under the Interfaces section and set-up the camera module in such a way that it can capture and send images as well as the capability to perform for live-streaming.
- **Coding:** Code the raspberry pi with a proper image processing and image classification model that is capable of capturing the images at regular intervals and then send it to the base station.

4.1.2 Hardware set-up for UAV

The set-up for the UAV involves the components that are required for the building of the UAV as well as the best components that need to be used to make the UAV fly better and for longer distances.

- **Quadcopter frame**: the quadcopter frame is the main base/skeleton of the UAV on which all the other hardware tools are placed.
- **Motors**: The second hardware component used here is a set of 4 motors with 2 clockwise and 2 anticlockwise motors that are placed on the quadcopter which can be used for aviation when a propeller is connected to it, for the convivence of the project we have decided to use 1200Kv BLDC motors for the movement of the UAV.
- **Electronic speed controller**: The next tool used is an electronic speed controller (ESC). It is one of the most important components of the drone as this component provides the power and the voltage required by the motors to turn the propellers.
- **Propellers**: Another important hardware component is a propeller that when given the proper amount of voltage and power can lift the drone into the air. We use 2 clockwise as well as 2 anticlockwise propellers for the proper working of the UAV.
- **Flight Controller:** This component is the main controller that controls the flight of the drone. This the most sensitive component of the drone and must be protected from external turbulence during the flight. Among the many flight controllers that were available, we have decided to use the KK2.1.5 Flight Controller which is cheap and at the same time has a better functionality when compared to the alternatives at the same price point.
- **Power Supply:** For powering the flight controllers and the motors we require lipo batteries carrying at least 3 cells of 2200mAh each.
- **Sensors for image capture**: Another component is a sensor that is used to capturing the images and then storing them before sending to the base station.

All these components were assembled and a small-scale prototype of an UAV was made as showed in figure 5.13.

4.2 Software Set-up

The software set-up mainly involves the writing the code, checking for any errors and the execution of the said code along with its compatibility on the raspberry pi. The steps involved in setting it up are as follows:

- **Update Raspberry Pi OS:** Open the terminal on the Raspberry Pi and run commands that can update the Raspberry Pi OS.
- **Install Camera Software:** Install the necessary software to interact with the camera module by running:
- **Python Script for Image Capture:** Create a Python script (e.g., **capture_image.py**) to interact with the camera and capture images. Use the Pi-Camera library in Python to control the Raspberry Pi camera module. Sample code for capturing an image.
- **Execute the Python Script:** Run the Python script from the terminal on the Raspberry Pi by navigating to the script location and executing. Adjust the script as needed for different settings such as image resolution, file format, file naming, etc.
- **Capturing Images at Intervals or Triggers:** Modify the Python script to capture images at regular intervals using loops or triggers (e.g., using GPIO pins for external triggers) for specific applications like time-lapse photography or event-based captures.
- **Transfer Images:** Once images are captured, they can be accessed directly on the Raspberry Pi or transferred to external storage or cloud services for further processing or storage.

Chapter 5 Results and Analysis

5.1 Image Processing:

After the dataset was trained by the many different algorithms as mentioned in chapter 3, the final best model was then decided based on some of the important criteria for choosing one such model are the classified into 2 main categories with the categories being Model complexity which indicates the computation time and the execution time of the model and the other one being Detection precision which brings out the accuracy of the model to predict the correct and the proper values. To evaluate and test out the latter, that is detection prediction we make use of metrics such as precision(P) and Recall(R) which are widely by many as parameters. Taking TP to represent the amount of True Positives samples, and FP to represent False Positive samples and FN to represent the number of False Negative samples. The mathematical expression for precision(P), recall(R) can be obtained through the following derivations:

Precision = $\frac{TP}{TP+FP}$……….(1)

$Re\ call$ = $\frac{TP}{TP+FN}$……….(2). Whereas the model complexity is determined by factors such as the dimensions of the model, the number of parameters it is processing and the amount of computational requirements. The larger the values indicate larger the complexity.

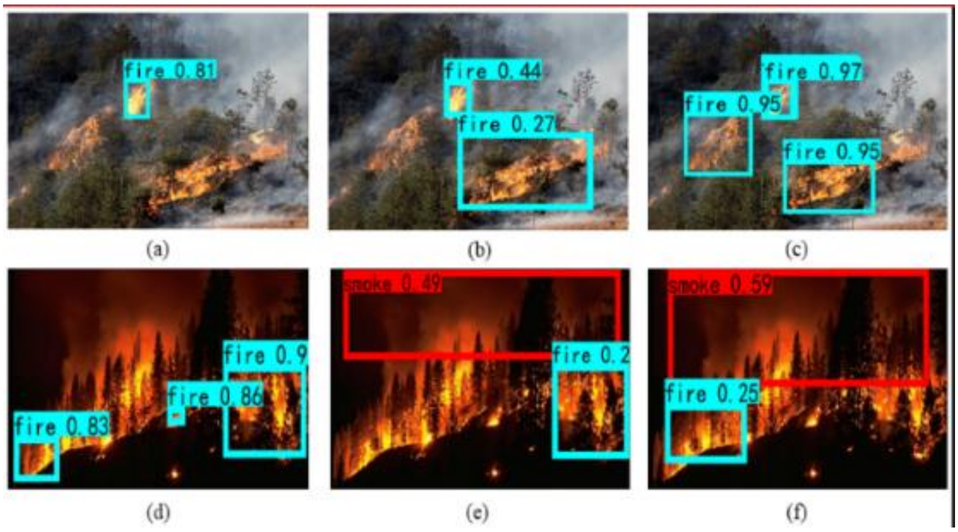

Fig 5.1: Images showing anchor boxes

The image shown above Figure 5.1 is a snippet of the machine learning algorithm being tested and validated where it’s predicting the images with fire and without fire, which is done in order to test if the model is predicting the images the correctly and to check if the model is trained properly with the dataset that it was supplied with.

The comparison table with IV of algorithms that we experimented with are shown below:

Table IV. Comparison between YOLO and SVM

Algorithm	Model Size/MB	Epochs used	Recall in %	Precision in %	Loss in %
YOLOv8	6.2	15	95	89.96	34.4
ReLU Activation function/SVM	3	5	-	-	29

The table above shows the comparison of the 2 such machine learning algorithms that we have experimented with and shows the main features that are being obtained starting from the size of the model to the loss percentage that is obtained with the YOLOv8 taking about 15 epochs per training while the ReLU /SVM algorithm trains using 5 epochs as SVM becomes unstable over large amounts of data.

From the table it is very evident that the YOLOv8 is a better choice when it comes to choosing an algorithm that can produce better results for the live-data while also taking a comparatively less time for charging the model thereby making it a more optimal model that can be used in order to detect and transmit the fire.

5.2 Graphs Obtained by the algorithm.

As mentioned in chapter 5.1 the loss and accuracy graph obtained form a major deciding factor in deciding which algorithm to be used and implemented.

So the accuracy and loss graphs that are obtained for the different algorithms is as shown below:

5.2.1 SVM/CNN algorithm.

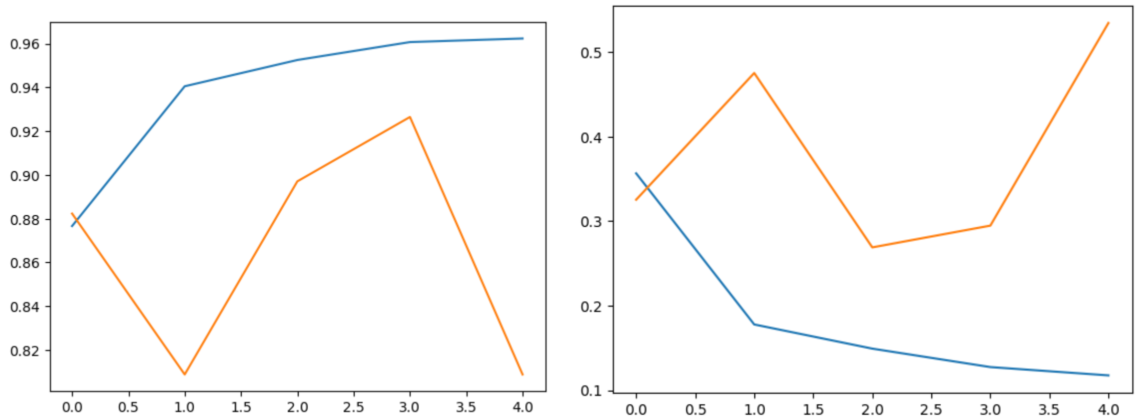

Fig 5.2: Loss and accuracy of the CNN model

As mentioned in the above figure 5.2, the graphs shown above shows the loss and the accuracy graphs of the CNN model obtained from testing and validating the model which shows how accurate the model's prediction is based on the results obtained.

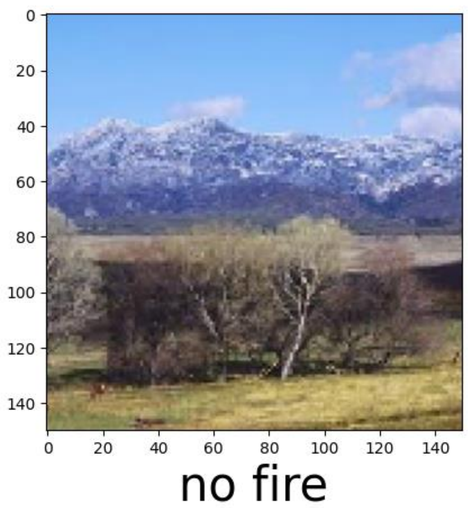

Fig 5.3: No Fire Output from the CNN model

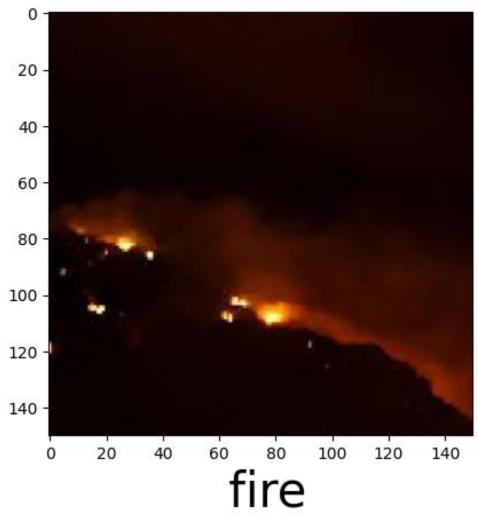

Fig 5.4: Fire Output from the CNN Model

The figures 5.3 and 5.4 shown above denote the model in action where it is being tested on a pre-defined image after the model has been trained completely.

5.2.2 YOLOv8 algorithm.

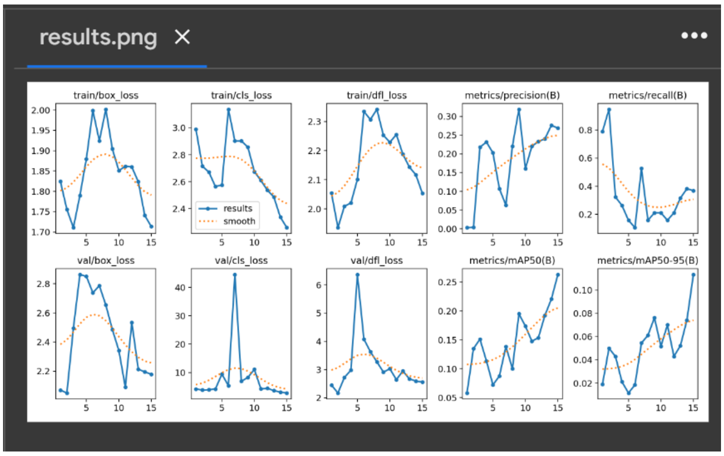

Fig 5.5: Graph showing the train validate model for different epochs

The figure 5.5 shown is the snippet of the image that shows the epochs being trained by the YOLOv8 model and the graphs associated with each of the epochs showing the accuracy and the preciseness of the data.

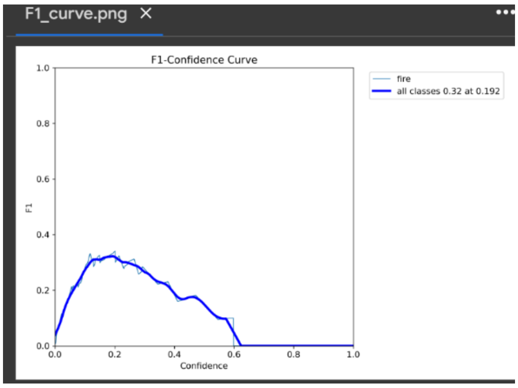

Fig 5.6: Graph showing F-1 Confidence Curve

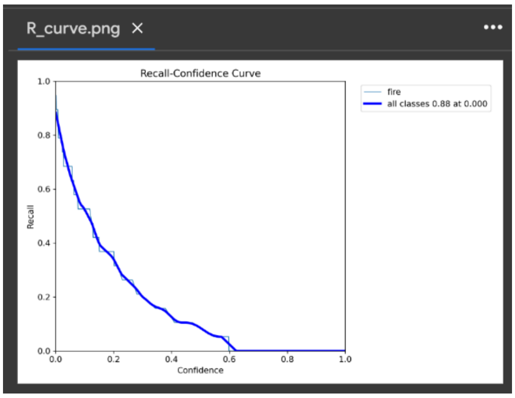

Fig 5.7: Recall Confidence Curve

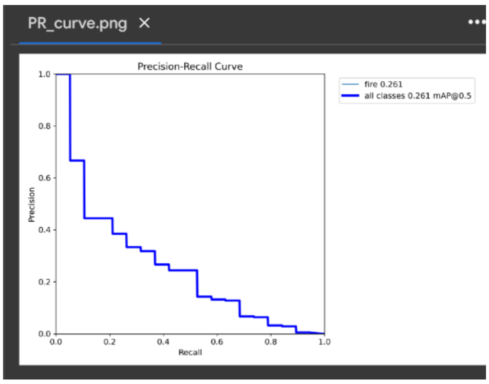

Fig 5.8: Precision Recall Curve

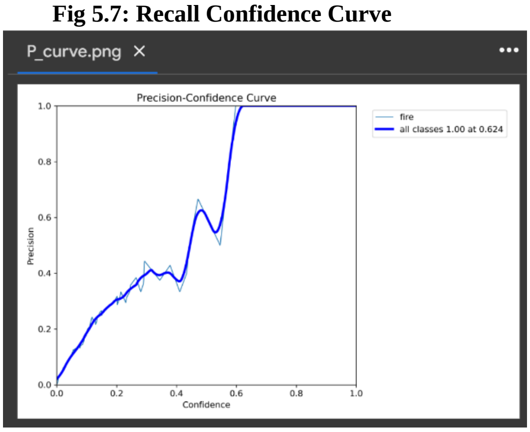

Fig 5.9: Precision Confidence Curve

The four images given above shows the four parameters obtained after training the model namely:

Accuracy: Accuracy is used to measure the performance of the model. It is the ratio of Total correct instances to the total instances shown in figure 5.8.

Precision: Precision is a measure of how accurate a model's positive predictions are. It is defined as the ratio of true positive predictions to the total number of positive predictions made by the model shown in figure 5.9.

Recall: Recall measures the effectiveness of a classification model in identifying all relevant instances from a dataset. It is the ratio of the number of true positive (TP) instances to the sum of true positive and false negative (FN) instances shown in figure 5.7.

F1-Score: F1-score is used to evaluate the overall performance of a classification model. It is the harmonic mean of precision and recall shown in figure 5.6.

All these four values combined together to give out a Confusion matrix which shows the complete and overall performance of the model and gives the rate at which the model predicts the correct values as well. The Confusion matrix obtained for the YOLOv8 algorithm is as shown below in figure.

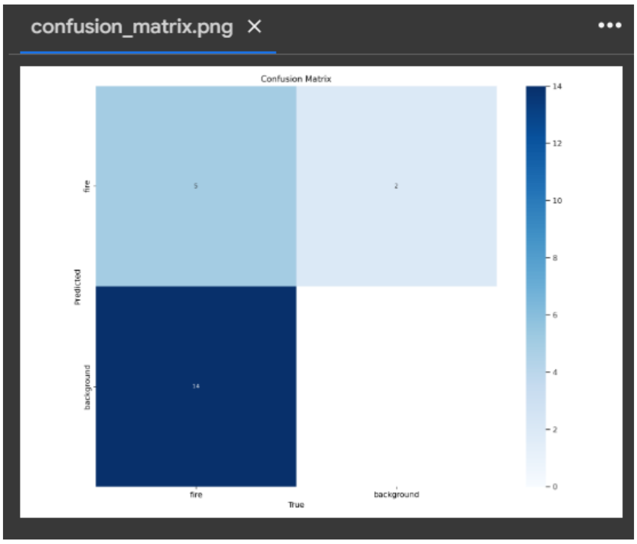

Fig 5.10: Confusion Matrix

The confusion matrix shown above in figure 5.10 gives the accurate representation of all the predictions made by the model and the rate at which the representation is made.

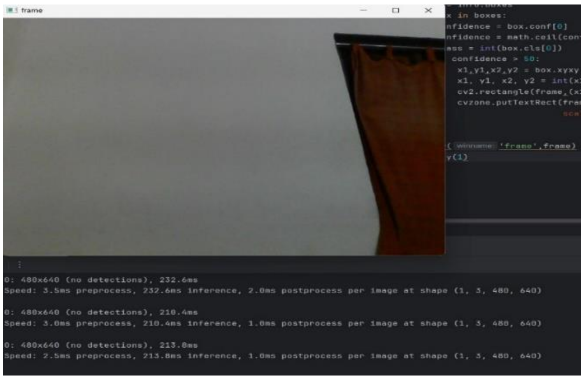

Fig 5.11: YOLO algorithm No Fire Output

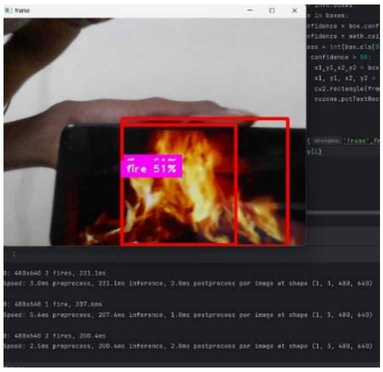

Fig 5.12: Yolo Algorithm Fire Output

The above results are the output obtained from the Machine Learning Algorithm using YOLO V5 are showned in figure 5.11 and 5.12. This model is highly efficient for object detection in real-time scenarios and provides high accuracy. But its speed is slightly reduced, but is good trade-off for providing better efficiency.

5.3 Prototype of the UAV

The other important aspect of the experiment was the building of the UAV which was supposed to fly and then help in capturing the images for the ML to predict.

The image showing the prototype of the UAV is as shown below in Figure 5.13:

Fig 5.13: Image showing the completed UAV

The above figure 5.13 shows the prototype of the UAV which is quadcopter being powered by a 3 cell 2200mAh battery with the flight controller that controls the entire operation of the UAV is a KK 2.1.5 Flight controller with the ESC providing the power to the Propellers and the motors which will give the thrust to the UAV so that it can take flight. After conducting the experiment and doing many trails with the UAV it was found that the UAV with a quadcopter frame is capable of carrying an additional payload of 1.5kg which can be improved by giving the quadcopter more power as well as increasing the number of propellers and evolving the quadcopter into a hexacopter and a octocopter which are capable of carrying more weight while at the same time using a better flight controller can account for a better stability during the flight time as well as better flight time. The ML model can also be improved to use a reenforcing learning algorithm which can adapt to changes in real-time based on the images fed into the model, and as the quadcopter is given more power and more weight-lifting capabilities, a small water propulsion system can be added to make it more useful for time-critical situations.

Chapter 6 Conclusion and Future Scope

We want to conclude our project report by stating that the growing no. of fires due to global warming and the number of accidental fires makes the fire fighting an important and crucial task and therefore by combing the power of the modern day technology along with some of the traditional methods it can be made possible to minimize the effect and spread of the fire as well as make the damage done to the fighters as well as the surroundings due to the very fire and therefore our project was intended to help the fire-fighters in minimizing the time needed by the rig to reach the centre of the fire as well as provide the fighters the pathway in order to rescue any victims that were trapped inside the fire.

The ways on how the project can be improved are as given the wide variety of applications and multiple applications and multiple degrees of freedom, it is challenging to design a natural interaction-based drone, and also control the UAV effectively. Especially when there is a lot of smoke or any obstacles blocking the visibility of the fire prone region, it is difficult for fire fighters to control the UAV. Making the UAV autonomous is the best way to solve this problem as the latest technologies support in making the UAV autonomous. The performance of UAV increases if its design is mainly for surveillance purposes, to detect source of the fire and share the information like apartment type, gas leakage, presence of living beings etc. to fire fighters. In addition to this, a feature can be added where if the UAV detects fire at initial stages itself, it can put down the fire using appropriate fire extinguishers which are carried by the drone using load carrying and releasing systems. Since fire is spread around less area correspondingly, it reduces the quantity of extinguishers required which is carried by the UAV. This will significantly reduce the payload size and weight so that a better load carrying and releasing system can be built to help the UAV to maneuver over the fire affected area with more stability. This is also a factor which will increase the overall efficiency and reliability of the UAV.

Below are few ways, that summarise on how our project can be updated and upgraded in order to obtain better results:

- Making the UAV Autonomous by using a processor with larger computing power as discussed above.

- Different sensors can be used on the UAVs to collect information about the environment, such as light intensity, pressure, wind speed, and direction.
- It is also possible to make the UAV completely ecofriendly by using solar technology with the technical advancements that make it achievable.
- This can be used for rescue operations along with detection and routing.
- Improved motor capacity with increased battery efficiency can improve the overall flight capacity.
- Better algorithms can be used for faster and more efficient detection of fire and smoke.
- Improved wing length which will enable more weight to be carried.

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Appendix – A

Appendix – B

Publication Details